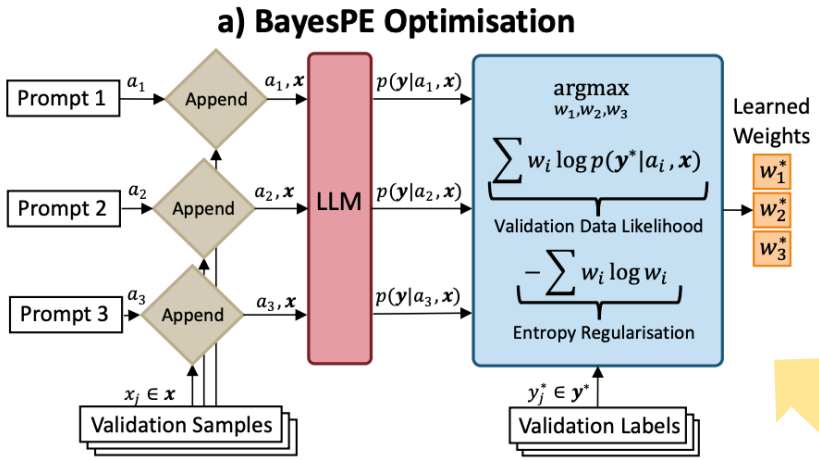
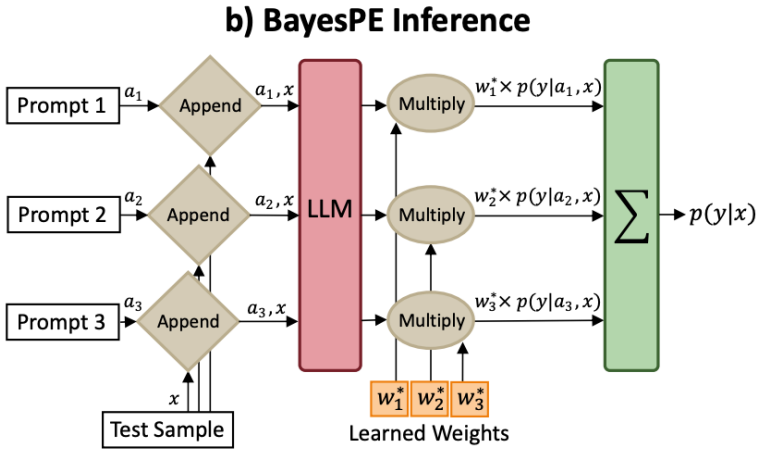
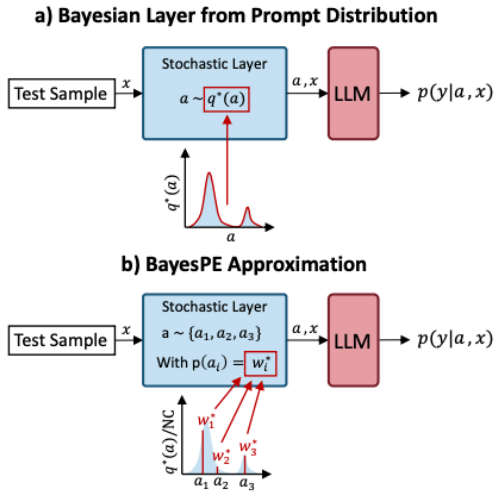
BayesPE

BayesPE[95]







$$\underset{\mathbf{w}}{\operatorname{argmax}} \underbrace{\sum_{i}^{N} w_i \log p(\mathbf{y}^*|a_i, \mathbf{x})}_{\text{Validation Data Likelihood}} \underbrace{- \sum_{i}^{N} w_i \log w_i}_{\text{Entropy}}$$

Loss

Regularization Term

☐ Pipeline:

(a) BayesPE optimisation: Each validation sample $x_j \in \mathbf{x}$ is appended to each semantically equivalent prompt a_i, and the LLM computes the likelihood $p(y_j^*|a_i, x_j)$ for the ground-truth label $y_j^* \in \mathbf{y}^*$; these likelihoods are used to learn prompt weights w_i^* via the BayesPE objective.

(b) BayesPE inference: For a test sample x, the LLM produces prompt-conditioned distributions $p(y|a_i, x)$, which are weighted by the learned weights w_i^* and aggregated to obtain the final distribution $p(y \mid x)$.

[95] Bayesian Prompt Ensembles: Model Uncertainty Estimation for Black-Box Large Language Models, ACL 2024

BayesPE[95]

- **Task Type:** Open-ended generation

- **Introduction:** Bayesian Prompt Ensembles (BayesPE) approximate Bayesian inference by sampling multiple prompt perturbations to induce posterior diversity in LLM outputs. It learns prompt-specific weights to aggregate multi-prompt predictions at inference time, yielding calibrated uncertainty estimates. BayesPE provides a lightweight, parameter-free Bayesian approximation for large language models.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Ensemble	Input	White-box	Post-hoc	Low	Calibration

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